

Work done when system goes from $A \rightarrow C$ (Change ABC) (18)

$$W_{A \rightarrow C} = \text{Area of curve ABC} \\ = \text{Area of } ABCC'A'A$$

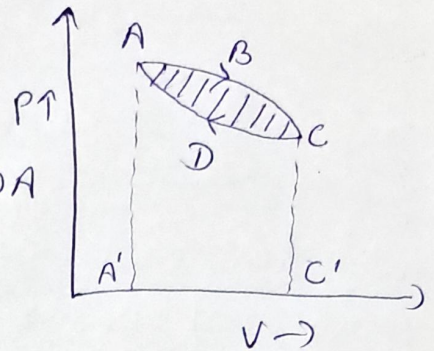


fig (4)

Similarly when system goes from $C \rightarrow A$ (Change CDA)

$$W_{C \rightarrow A} = \text{Area of curve CDA} \\ = \text{Area of } CDAA'C'C$$

$\therefore$ Net work done during complete cycle

$$W = W_{A \rightarrow C} - W_{C \rightarrow A}$$

$$= \text{Area } ABCC'A'A - \text{Area } CDAA'C'C$$

$$W = \text{Area ABCDA} \quad (\text{Shaded region in fig (4)})$$

Thus, work done during a cyclic process is equal to the area enclosed by its indicator diagram in the P-V graph.

During a cyclic change is traced in clockwise direction then the net amount of work done is positive and if it is traced in anti-clockwise direction then net amount of work done is negative.

Internal energy

The energy possessed by a body or a system by virtue of its molecular arrangement and motion of molecules is called internal energy or

or ~~intrinsic~~ intrinsic energy or microscopic energy. This energy is associated with the thermodynamic state of the system. (16)

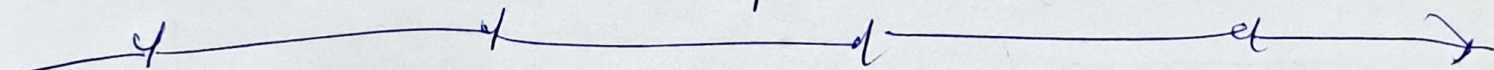
For perfect gases internal energy is a function of temperature only i.e. $U(T)$ as molecular forces of attraction and repulsion are absent. For real gases, vapour or liquid the internal energy strongly depends upon temperature and pressure i.e. $U(T, P)$

Characteristics of internal energy

- 1) Internal energy is associated with the thermodynamic state of the system. Thus it is a point function and a thermodynamic property.
- 2) It is an extensive property, ^{as it} and depends upon the mass of the system.

3) Also, $dU = m C_v dT$

$$\text{or } U_2 - U_1 = m \int_{T_1}^{T_2} C_v dT = m C_v (T_2 - T_1)$$



First law of Thermodynamics

Heat is a form of energy and energy is always conserved.

First law of thermodynamics is the statement of the principle of conservation of energy. It states that net flow of energy across the boundary of a system is equal to the change in energy of the system.

Let dQ amount of heat be supplied to a system. Then it is utilized in three ways by the system :-

- i) Partially, it is utilized in raising the temperature of the body which is equivalent of increasing its internal kinetic energy (U_K).
- ii) A part of it is utilized in doing the external work against molecular attractions which is equivalent of increasing the internal potential energy of the system (U_P).
- iii) The remaining heat energy is utilized in expanding the system doing external work.

If dU_K and dU_P are the changes in internal kinetic energy and internal potential energy and dW is the external work done then

$$dQ = dU_K + dU_P + dW$$

or $dU_K + dU_P = dU$ i.e. increase in total internal energy of the system (18)

$$\therefore \boxed{dQ = dU + dW}$$

This is differential form of first law of thermodynamics and can also be stated as:

In all transformations the heat energy supplied must be balanced by the external work done plus the ~~increase~~^{change} in internal energy.

→ Corollaries of First law of Thermodynamics

i) Internal energy of the universe is constant

By first law of thermodynamics, $dQ = dU + dW$ (1)
where dQ is the amount of heat absorbed by the system and performs work dW and at the same time dU is the change in internal energy.

For surroundings dQ amount of heat is lost by it and dW work is done on it. If dU' is the change in its internal energy then for surroundings

First law of thermodynamics is

$$-dQ = dU' - dW$$

$$\text{or } dU' = dW - dQ \quad \text{--- (2)}$$

eqn (1) can be written as

$$dU = dQ - dW$$

$$\text{or } dU = -(dW - dQ) \quad \text{--- (3)}$$

using eqn (3) in eqn (2)

$$dU' = -dU$$

$$\text{or } dU' + dU = 0$$

$$\Rightarrow \boxed{U' + U = \text{constant}}$$

Thus, net change in internal energy of the system and surroundings taken together would be zero.

Applications of First law of thermodynamics

1) Isothermal Process

When a change in pressure and volume of a substance takes place but the temperature remains constant, the change is said to be isothermal.

Since temperature remains constant, thus there will be no change in the internal energy (U) since U is a function of T i.e. $dU = 0$ — (1)

According to first law of thermodynamics

$$dQ = dU + dW \text{ — (2)}$$

Using equ. (1) in equ. (2)

$$\boxed{dQ = dW}$$

Thus during ~~isothermal~~ isothermal process, whatever

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amount of heat is supplied to the system, it is utilized in doing the external work.

2) Adiabatic Process

Adiabatic Process is the one in which no heat is allowed ~~or~~ ~~to~~ enter or leave the system, however its pressure and volume change.

$$\Rightarrow dQ = 0. \text{ --- (1)}$$

According to 1st law of thermodynamics:

$$dQ = dU + dW \text{ --- (2)}$$

using eqn (1) in eqn (2)

$$0 = dU + dW$$

$$\boxed{dU = -dW}$$

Thus, in ~~isobaric~~ adiabatic process whatever work is done whether expansion or compression, it is done at the cost of internal energy.

3) Isobaric Process

During this process pressure of the thermodynamic system remains constant.

According to 1st law of thermodynamics

$$dQ = dU + dW$$

Thus, during isobaric process whatever amount of heat is supplied to the system, a part of it is used

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$(dW = PdV)$

in doing the external work, and rest is used in changing the internal energy of the system.

4) Isochoric Process

A thermodynamic process during which the volume of the system remains constant is called isochoric process.

According to 1st law of thermodynamics

$$dQ = dU + dW \quad \text{--- (1)}$$

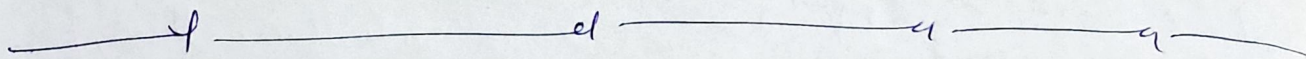
Since, here volume remains constant or ~~is~~ unchanged $\Rightarrow dV = 0$

$$\therefore \text{work done } dW = PdV = 0 \quad \text{--- (2)}$$

Using equ. (2) in equ. (1)

$$\boxed{dQ = dU}$$

Thus, during an isochoric process whatever amount of heat is supplied or absorbed from the system is utilized in changing its internal energy.



Limitations of 1st law of thermodynamics

- i) This law does not provide any information about the direction of energy transformation.
- ii) It does not consider the grade of energy or energy ~~reservoir~~ reservoirs. It considers all energy reservoirs to be identical.
- iii) It considers all forms of energy to be equivalent i.e. it does not tell us the ~~quantum~~ quantum of energy which will change one form to the other.

Reversible Processes

A reversible process is the one that can be retraced in opposite direction so that the working substance passes through exactly the same condition as it does in the direct process.

Conditions to be fulfilled for a reversible process

- i) System undergoing such a change should not lose heat by conduction, convection or radiation. Thus, for a process to be reversible it should be devoid of any dissipative effects like ~~friction~~ friction, electrical

Resistance etc.

ii) For a process to be reversible, it should take place infinitesimally slowly.

These conditions can ~~be~~ never be met strictly because friction can never be absent ~~no~~ and also no material can be a perfect conductor or insulator.

Examples of reversible process

- 1) Ice melts when certain amount of heat is absorbed by it and the water so formed can be converted back to ice if same amount of heat is removed from it.
- 2) electrolysis.
- 3) Isothermal and adiabatic changes carried out at infinitesimal slow rates.

Irreversible Process

~~During an irreversible process the~~

Irreversible process cannot be retraced in the opposite direction i.e. the working substance does not pass through the same intermediate states that it passed during the direct process.

Examples of Irreversible process

- ① Work done ~~again~~ against friction.
- ② Heat produced during the passage of electric current through a wire or any ~~se~~ electrical circuit.
- ③ Rusting of iron.

Heat engine and Thermal efficiency

Heat engine is a device which converts heat energy into work.

Heat engine comprises of the following main components:

1) Source/heat reservoir:-

A source is a hot body maintained at a very high fixed temperature (T_1) from ~~to~~ which heat engine can draw any amount of heat.

2) Sink/cold reservoir:-

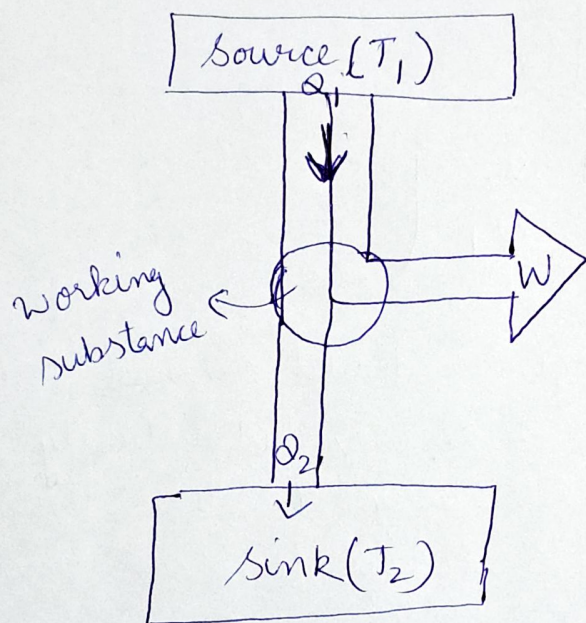
A sink is a cold body maintained at a very low fixed temperature (T_2) to which heat engine can reject any amount of heat.

3) Working substance:-

Working substance is the material which ~~on~~

When supplied with ~~sa~~ heat energy performs the mechanical work. For examples, in case of automobiles that are fitted with internal-combustion engine ~~contains~~ contains the mixture of air and fuel which act as working substance, in case of steam engine this role is played by water.

A schematic diagram of heat engine showing flow of heat energy in it is shown below.



The working substance in the heat engine absorbs ~~Q1~~ heat from the source converts a part of it into external work W and gives rest of the heat to sink Q_2 and acquires initial state. The similar work can be obtained by performing the same cycle again.

The useful output of engine is the net work done (W) by the working substance

$$\therefore W = Q_1 - Q_2 \quad \text{--- (1)}$$

The efficiency of heat engine is defined as the ratio of net work done to the ~~total~~ input heat energy and is denoted by η .

$$\therefore \boxed{\eta = \frac{W}{Q_1}} \quad \text{--- (2)}$$

or using equ. (1) in equ. (2)

$$\boxed{\eta = \frac{Q_1 - Q_2}{Q_1}}$$

or ~~η~~ $\boxed{\eta = 1 - \frac{Q_2}{Q_1}}$

~~Let~~ Ideally a heat engine should convert whole heat Q_1 into external work i.e. $Q_2 = 0$, but actually this is impossible, some heat is always wasted, thus Q_2 can never be zero $\Rightarrow \eta$ is always less than unity ($\because$ if $Q_2 = 0$, then $\eta = 1$)

Thus a heat engine is ~~highly~~ highly efficient if it converts ~~most~~ of minimum amount of absorbed heat to useful work and rejects minimum possible Q_2 to sink. $\Rightarrow$ smaller the Q_2 efficient will be the heat engine.

————— η ————— η ————— η ————— η —————

Carnot's cycle

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A ~~car~~ Carnot cycle is a hypothetical cycle consisting of four distinct processes i.e. two reversible isothermal processes and two reversible adiabatic processes.

The essential elements required for this cycle to take place:-

- i) A working substance is considered to be a perfect gas.
- ii) Two heat reservoirs - one is source at high temperature (T_1) and other is sink at low temperature (T_2).
- iii) Piston-cylinder assembly in order to get work out of the working substance. The walls of cylinder and piston are considered to be perfect heat insulators. The cylinder head can be provided with a diathermic cover (perfect heat conductor) and an adiabatic cover (perfect heat insulator) alternatively. Also there should be no friction when piston moves inside cylinder.

The experimental set up required for this cycle to take place is shown ~~as imp~~ ~~for~~ in fig (1) as.

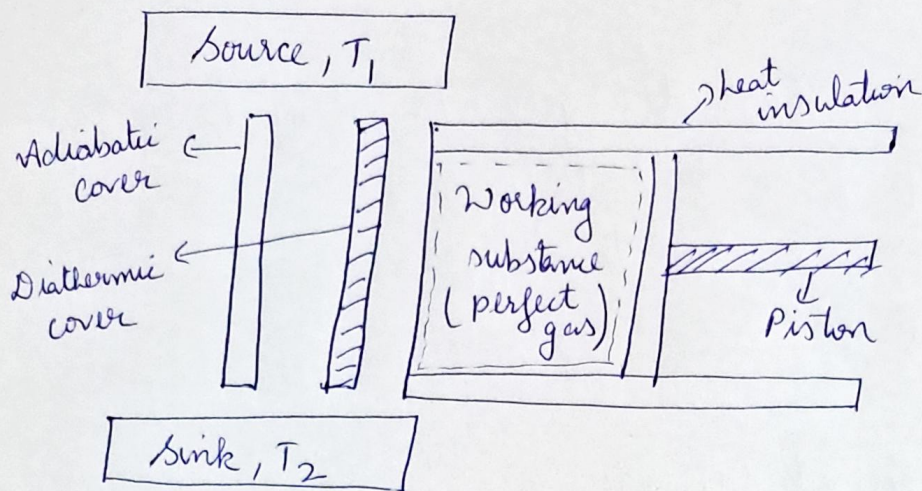


fig. (1)

The different processes taking place in a Carnot's cycle is represented on PV -graph and is shown in fig (2)

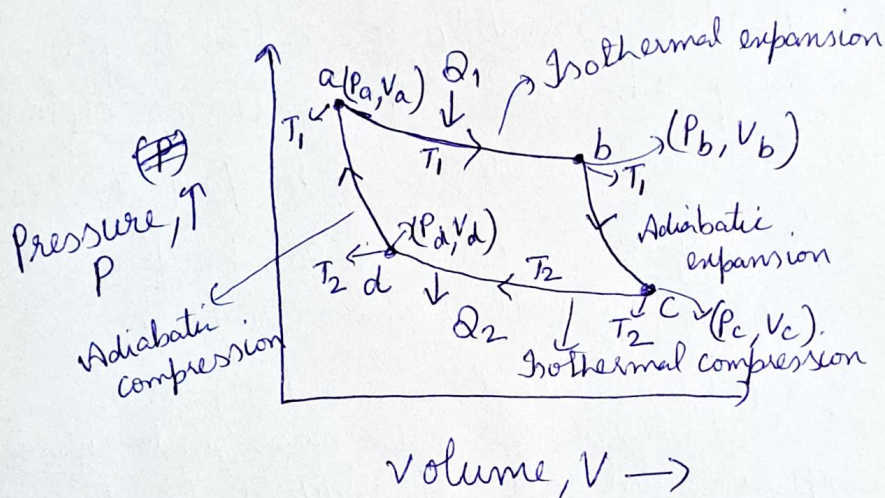


fig. (2)

i) Isothermal expansion (a $\rightarrow$ b) :-

This process is carried out at constant temperature T_1 and Q_1 is the amount of heat supplied during this

process. This process is carried out by bringing the cylinder head in thermal contact with the source through diathermic cover. The gas undergoes isothermal expansion from point a (P_a, V_a) to point b (P_b, V_b). The work done during ~~pro~~ this process is ~~same~~ equivalent to the heat supplied. i.e.

$$W_{a \rightarrow b} = Q_1 = nRT_1 \log_e \frac{V_b}{V_a} \quad \text{--- (1)}$$

[where n is no. of moles of gas]

Adiabatic expansion (b $\rightarrow$ c)

When the isothermal expansion ended at point b then the diathermic cover at cylinder head is replaced by the adiabatic cover and the gas is allowed to expand adiabatically and ~~again again~~ attains state c (P_c, V_c). During this process no heat was supplied to the gas and the work is done at the cost of its internal energy and finally its temperature falls to T_2 at point c.

Isothermal Compression (c $\rightarrow$ d)

At point c the cylinder head is ~~in thermal~~ is again ~~is again~~ replaced with the diathermic cover and is placed in thermal contact with the sink and gas is allowed to compress isothermally at temperature T_2 upto point d. During this process the gas rejects Q_2 amount of heat to the sink. By sign convention

Q_2 must be negative. Thus work done during this process ⁽³⁰⁾ is equivalent to amount of heat rejected by the gas i.e.

$$W_{c \rightarrow d} = -Q_2 = nRT_2 \log_e \frac{V_d}{V_c}$$

$$\text{or } W_{c \rightarrow d} = Q_2 = nRT_2 \log_e \frac{V_c}{V_d} \quad \text{--- (2)}$$

Adiabatic compression (d $\rightarrow$ a)

When the isothermal compression ends at point B, the cylinder head is replaced by adiabatic cover and the gas is allowed to compress adiabatically till it acquires the initial point a. During this process the work is done on the gas as a result its internal energy increases and finally at point a it attains temperature T_1 .

A heat engine working on Carnot cycle is called Carnot's engine and its thermal efficiency is

$$\eta = \frac{\text{Net work done}}{\text{Input heat}}$$

$$\eta = \frac{Q_1 - Q_2}{Q_1} \quad \text{--- (3)}$$

Using equ. (1) and equ. (2) in equ. (3)

$$\eta = \frac{nRT_1 \log_e \frac{V_b}{V_a} - nRT_2 \log_e \frac{V_c}{V_d}}{nRT_1 \log_e \frac{V_b}{V_a}}$$

or

$$\eta = 1 - \frac{T_2}{T_1} \frac{\log_e \frac{V_c}{V_d}}{\log_e \frac{V_b}{V_a}} \quad \text{--- (4)}$$

For adiabatic expansion $b \rightarrow c$

$$T_b V_b^{\gamma-1} = T_c V_c^{\gamma-1}$$

$$\text{or } \frac{T_b}{T_c} = \left(\frac{V_c}{V_b} \right)^{\gamma-1}$$

[where γ is the ratio of specific heat at constant pressure to specific heat at constant volume]

Also, $T_b = T_1$ and $T_c = T_2$

$\therefore$ Above relation becomes

$$\frac{T_1}{T_2} = \left(\frac{V_c}{V_b} \right)^{\gamma-1} \quad \text{--- (5)}$$

For adiabatic compression $d \rightarrow a$

$$T_d V_d^{\gamma-1} = T_a V_a^{\gamma-1}$$

$$\text{or } \frac{T_d}{T_a} = \left(\frac{V_a}{V_d} \right)^{\gamma-1}$$

Also, $T_d = T_2$ and $T_a = T_1$

$\therefore$ ~~Above~~ Above relation becomes

$$\frac{T_2}{T_1} = \left(\frac{V_a}{V_d} \right)^{\gamma-1}$$

$$\text{or } \frac{T_1}{T_2} = \left(\frac{V_d}{V_a} \right)^{\gamma-1} \quad \text{--- (6)}$$

Equating eqn. (5) and eqn. (6)

$$\left(\frac{V_c}{V_b} \right)^{\gamma-1} = \left(\frac{V_d}{V_a} \right)^{\gamma-1}$$

$$\frac{V_c}{V_b} = \frac{V_d}{V_a} \quad \text{or } \frac{V_c}{V_d} = \frac{V_b}{V_a} \quad \text{--- (7)}$$

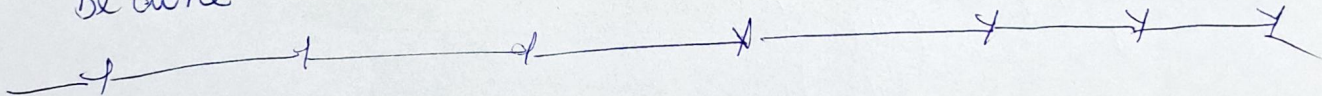
Using equ (7) in equ (4)

$$\boxed{\eta = 1 - \frac{T_2}{T_1}} \quad \text{--- (8)}$$

$$\text{or } \boxed{\eta = \frac{T_1 - T_2}{T_1}} \quad \text{--- (9)}$$

Conclusions that can be drawn from efficiency of Carnot's engine :

- i) The efficiency is independent of the working substance and depends only upon the temperature of the ~~work~~ source and sink.
- ii) The efficiency is directly proportional to the temperature difference ($T_1 - T_2$) of source and sink. Higher the temperature difference higher the efficiency or hotter the source and colder is the sink higher will be the efficiency.
- iii) If $T_2 = 0$ then η becomes 100%. This implies absence of sink which is the violation of 2nd law of thermodynamics.
- iv) If $T_1 = T_2$ then η will be zero $\Rightarrow$ both sink and source at same temperature and hence no work will be done.



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Carnot's cycle is impractical or practically impossible.

- i) All the four processes should be reversible $\Rightarrow$ there should be no forces of interaction among the gas molecules and no mechanical friction between piston and cylinder. This is practically impossible.
- ii) There is no much variation in the slopes of isothermal and adiabatic lines $\Rightarrow$ P-V plot is extended greatly both in horizontal and vertical directions. Thus the cylinder involves very high pressure and large volumes which makes it bulky and heavy.
- iii) ~~Isothermal~~ During isothermal and adiabatic processes the piston is to be moved at different speeds which is very difficult to manage practically.

